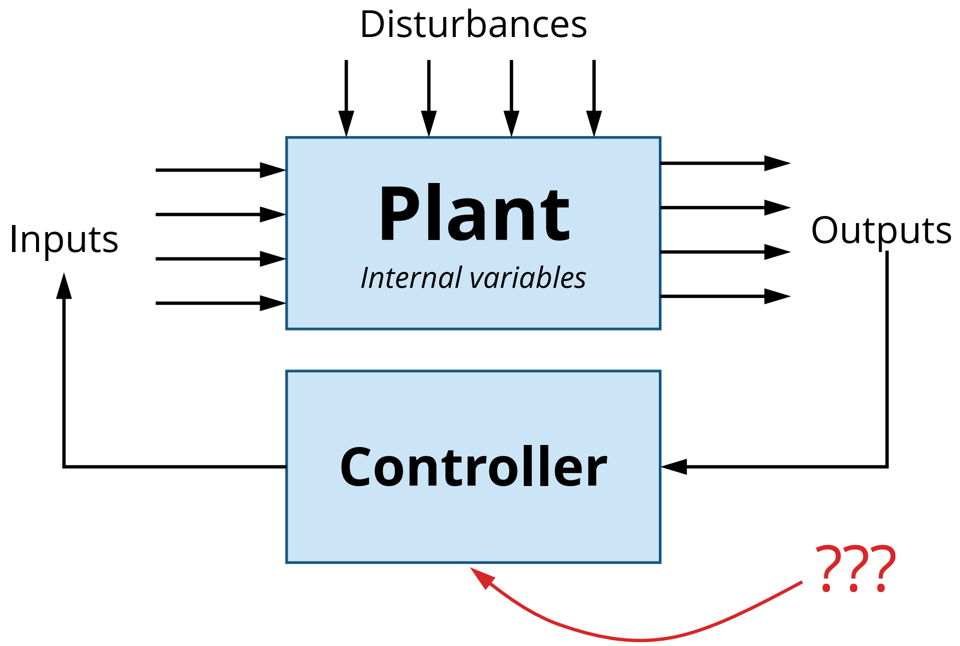

AA 513 Multivariable Control

Lecture 06 // Week 6

Instructor: Prof. Karen Leung

Announcements

- Exercises 9, 10, 11, 12 out (soon)
- Self-assessments due this week
- Check-in sessions by appointment to discuss research papers and tutorial topics.
- Research paper presentation
  - Follow PACES framework
  - Shared slide deck released
- Tutorial submission
  - **Submission**: Upload video to shared folder
  - **Peer review**: You will be assigned videos to review. You are welcome to review more if you like.

Week 10 schedule

Schedule

- 6 minutes per talk + 2 minute questions

Session 1			Session 2			Session 3		
#	Time	Name	#	Time	Name	#	Time	Name
1	6:00PM		8	7:00PM		15	8:00PM	
2	6:08PM		9	7:08PM		16	8:08PM	
3	6:16PM		10	7:16PM		17	8:16PM	
4	6:24PM		11	7:24PM		18	8:24PM	
5	6:30PM		12	7:30PM		19	8:30PM	
6	6:38PM		13	7:38PM		20	8:38PM	
7	6:46PM		14	7:46PM		21	8:46PM	
						22	8:54PM	

Last time

- Guest lecture by Dr Liam Krause, topic on Markov Decision Process and algorithms for finding optimal policies
  - **MDP**: More general framework for modeling systems that evolve over time
    - Assumes transition *probabilities*
    - Expectation over transition probabilities
  - **Value iteration**: DP until convergence
  - **Policy iteration**: Iterate between policy evaluation & policy improvement
  - **POMDPs**: *Partially observable* MDPs. Keep a *belief* over your state.
    - Expectation over transition AND belief probabilities
  - **Street-fighting techniques**: Value function approximations, MCTS (local search)

This week

- Continuous time: Hamilton-Jacobi-Bellman equation
- Linear quadratic regulator
- No lecture next week (Veteran’s Day)
  - Will release recording

Exercises

- Ex9 Stochastic Bellman equation
- Ex10: Finite horizon LQR
- Ex11: Infinite horizon LQR
- Ex12: Tracking LQR

Continuous-time, continuous state & control

- Consider cost $g_N(\boldsymbol{x}(T)) + \int_0^T g(x(t), u(t), t)dt$
- Cast as discrete time problem with timestep ϵ
  - $g_{discrete}(x_t, u_t, t) = \int_t^{t+\epsilon} g(x(\tau), u(\tau), \tau)d\tau \approx \epsilon g(x(t), u(t), t)$, for small ϵ

$$V^*(\boldsymbol{x}(t), t) = \min_{\boldsymbol{u}(t)\in U(\boldsymbol{x}(t))} \epsilon g(\boldsymbol{x}(t), \boldsymbol{u}(t), t) + V^*(\boldsymbol{x}(t+\epsilon), t+\epsilon)$$

$$0 = \min_{\boldsymbol{u}(t)\in U(\boldsymbol{x}(t))} \epsilon g(\boldsymbol{x}(t), \boldsymbol{u}(t), t) + V^*(\boldsymbol{x}(t+\epsilon), t+\epsilon) - V^*(\boldsymbol{x}(t), t)$$

$$0 = \min_{\boldsymbol{u}(t)\in U(\boldsymbol{x}(t))} \epsilon g(\boldsymbol{x}(t), \boldsymbol{u}(t), t) + (V^*(\boldsymbol{x}(t), t) + \nabla_x V^*(\boldsymbol{x}(t), t)^T f(x(t), u(t), t)\epsilon + \frac{\partial V}{\partial t}\epsilon + O(\epsilon^2) - V^*(\boldsymbol{x}(t), t)$$

$$0 = \min_{\boldsymbol{u}(t)\in U(\boldsymbol{x}(t))} g(\boldsymbol{x}(t), \boldsymbol{u}(t), t) + \frac{1}{\epsilon}(\nabla_x V^*(\boldsymbol{x}(t), t)^T f(x(t), u(t), t)\epsilon + \frac{\partial V}{\partial t}\epsilon + O(\epsilon^2))$$

$$0 = \lim_{\epsilon\to 0} \min_{\boldsymbol{u}(t)\in U(\boldsymbol{x}(t))} g(\boldsymbol{x}(t), \boldsymbol{u}(t), t) + \nabla_x V^*(\boldsymbol{x}(t), t)^T f(x(t), u(t), t) + \frac{\partial V}{\partial t} + O(\epsilon)$$

$$0 = \frac{\partial V}{\partial t}(\boldsymbol{x}(t), t) + \min_{\boldsymbol{u}(t)\in U(\boldsymbol{x}(t))} g(\boldsymbol{x}(t), \boldsymbol{u}(t), t) + \nabla V(\boldsymbol{x}(t), t)^T f(\boldsymbol{x}(t), \boldsymbol{u}(t), t)$$

Hamilton-Jacobi-Bellman equation

Consider cost $g_N(\boldsymbol{x}(T)) + \int_0^T g(\boldsymbol{x}(t), \boldsymbol{u}(t), t)dt$

$$\frac{\partial V}{\partial t}(\boldsymbol{x}(t), t) + \min_{\boldsymbol{u}(t) \in U(\boldsymbol{x}(t))} \{g(\boldsymbol{x}(t), \boldsymbol{u}(t), t) + \nabla V(\boldsymbol{x}(t), t)^T f(\boldsymbol{x}(t), \boldsymbol{u}(t), t)\} = 0$$

Terminal condition: $V(\boldsymbol{x}, T) = g_N(\boldsymbol{x})$

Solve PDE backward in time. But generally not easy to solve ☹

What have we just done?

- Derived the key equations describing the optimal solution to a sequential decision-making problem
  - Under some assumptions, these are necessary and sufficient conditions for optimality

- Solving exactly is hard / intractable
- Instead, perform "**approximate** dynamic programming" and **function approximators** for value function

Can we ever hope to solve these equations exactly?!

- Yes! Under strict assumptions on the functional form of dynamics, cost, and value function.
- This is known as the linear quadratic regulator!

The discrete-time finite horizon LQR problem

Regulate to zero, want state and control to be small

$$\min_{\mathbf{u}_{0:T-1}} \quad \mathbf{x}_T Q_T \mathbf{x}_T + \sum_{t=0}^{T-1} \mathbf{x}_t Q_t \mathbf{x}_t + \mathbf{u}_t R_t \mathbf{u}_t \tag{1}$$

Quadratic cost

$$\text{subject to} \quad \mathbf{x}_{t+1} = A_t \mathbf{x}_t + B_t \mathbf{u}_t, \quad t = 0, \ldots, T-1 \tag{2}$$

Linear dynamics

$$\mathbf{x}_0 = \mathbf{x}_{\text{current}} \tag{3}$$

Initial state

* Not really a constraint since state is not an optimization variable. More to just “anchor” the problem to a starting state.

Some notes about the LQR problem

$$\begin{aligned} \min_{\mathbf{u}_{0:T-1}} \quad & \mathbf{x}_T Q_T \mathbf{x}_T + \sum_{t=0}^{T-1} \mathbf{x}_t Q_t \mathbf{x}_t + \mathbf{u}_t R_t \mathbf{u}_t \\ \text{subject to} \quad & \mathbf{x}_{t+1} = A_t \mathbf{x}_t + B_t \mathbf{u}_t, \quad t = 0, \dots, T-1 \\ & \mathbf{x}_0 = \mathbf{x}_{\text{current}} \end{aligned}$$

- No constraints! (aside from dynamics)
- Cost and dynamics can be ***time-varying***
- Is it always possible to regulate to zero? (i.e., stabilize system?)
  - Can we even control it?
  - What if there are uncontrollable/unobservable modes?
- Conditions on the cost matrices.

How to solve the LQR problem?

$$
\begin{aligned}
\min_{\mathbf{u}_{0:T-1}} \quad & \mathbf{x}_T Q_T \mathbf{x}_T + \sum_{t=0}^{T-1} \mathbf{x}_t Q_t \mathbf{x}_t + \mathbf{u}_t R_t \mathbf{u}_t \\
\text{subject to} \quad & \mathbf{x}_{t+1} = A_t \mathbf{x}_t + B_t \mathbf{u}_t, \quad t = 0, \ldots, T-1 \\
& \mathbf{x}_0 = \mathbf{x}_{\text{current}}
\end{aligned}
$$

- Is the problem convex?
- Can we solve it in closed-form?

Recall: Bellman equation

$$V^\star(\mathbf{x}_t, t) = \min_{\mathbf{u}\in\mathcal{U}(\mathbf{x}_t)} g(\mathbf{x}_t, \mathbf{u}_t, t) + V^\star(f(\mathbf{x}_t, \mathbf{u}_t, t), t+1), \; V^\star(\mathbf{x}_T, T) = g_T(\mathbf{x}_T)$$

$$g_T(\mathbf{x}_T)$$

$$f(\mathbf{x}_t, \mathbf{u}_t, t)$$

$$g(\mathbf{x}_t, \mathbf{u}_t, t)$$

$$\mathbf{u}\in\mathcal{U}(\mathbf{x}_t)$$

Simplifying assumptions: Not time-varying

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Discrete-time finite horizon LQR (time-invariant)

1. set $P_T = Q_T$
2. for $t = N, ..., 1,$

$$P_{t-1} = Q + A^T P_t A - A^T P_t B(R + B^T P_t B)^{-1} B^T P_t A$$

$$K_{t-1} = -(R + B^T P_t B)^{-1} B^T P_t A$$

3. for $t = 0, \dots, N-1$, $\mathbf{u}_t^\star = K_t \mathbf{x}_t$

Discrete-time finite horizon LQR (time-varying)

What changes?

1. set $P_T = Q_T$

2. for $t = N, ..., 1,$

$$P_{t-1} = Q + A^T P_t A - A^T P_t B(R + B^T P_t B)^{-1} B^T P_t A$$

$$K_{t-1} = -(R + B^T P_t B)^{-1} B^T P_t A$$

3. for $t = 0, \ldots, N-1$, $\mathbf{u}_t^\star = K_t \mathbf{x}_t$

Discrete-time infinite horizon LQR (time-invariant)

What changes?

1. set $P_T = Q_T$

2. for $t = N, ..., 1,$

$$P_{t-1} = Q + A^T P_t A - A^T P_t B(R + B^T P_t B)^{-1} B^T P_t A$$

$$K_{t-1} = -(R + B^T P_t B)^{-1} B^T P_t A$$

3. for $t = 0, \ldots, N-1$, $\mathbf{u}_t^\star = K_t \mathbf{x}_t$

What about the continuous-time setting?

- Same process but with the HJB equation!

$$0 = \frac{\partial V}{\partial t}(\mathbf{x}(t), t) + \min_{\mathbf{u}(t) \in \mathcal{U}(\mathbf{x}(t))} g(\mathbf{x}(t), \mathbf{u}(t), t) + \nabla V(\mathbf{x}(t), t)^T f(\mathbf{x}(t), \mathbf{u}(t), t)$$

$$\begin{aligned}
\min_{\mathbf{u}(\cdot)} \quad & \mathbf{x}(T) Q_T \mathbf{x}(T) + \int_0^T \mathbf{x}(t) Q(t) \mathbf{x}(t) + \mathbf{u}(t) R(t) \mathbf{u}(t) && (1) \\
\text{subject to} \quad & \dot{\mathbf{x}} = A(t)\mathbf{x}(t) + B(t)\mathbf{u}(t), \quad t = [0, T] && (2) \\
& \mathbf{x}(0) = \mathbf{x}_{\text{current}} && (3)
\end{aligned}$$

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Continuous-time finite horizon LQR (time-invariant)

1. set $P(T) = Q_T$
2. Solve Riccati differential equation *backward* in time,

$$-\dot{P}(t) = A^T P(t) + P(t)A - P(t)BR^{-1}B^T P(t) + Q, \quad P(T) = Q_T$$

3. Optimal control $\mathbf{u}(t)^\star = K(t)\mathbf{x}(t)$, $K(t) = -R^{-1}B^T P(t)$

Continuous-time finite horizon LQR (time-varying)

What changes?

1. set $P(T) = Q_T$

2. Solve Riccati differential equation *backward* in time,

$$-\dot{P}(t) = A^T P(t) + P(t)A - P(t)BR^{-1}B^T P(t) + Q, \quad P(T) = Q_T$$

3. Optimal control $\mathbf{u}(t)^\star = K(t)\mathbf{x}(t)$, $K(t) = -R^{-1}B^T P(t)$

Continuous-time infinite horizon LQR (time-invariant)

What changes?

1. set $P(T) = Q_T$

2. Solve Riccati differential equation *backward* in time,

$$-\dot{P}(t) = A^T P(t) + P(t)A - P(t)BR^{-1}B^T P(t) + Q, \quad P(T) = Q_T$$

3. Optimal control $\mathbf{u}(t)^\star = K(t)\mathbf{x}(t)$, $K(t) = -R^{-1}B^T P(t)$

Other variations

- Regulating to $\left(\boldsymbol{x}-\boldsymbol{x}_g\right)^T Q\left(\boldsymbol{x}-\boldsymbol{x}_g\right)+\left(\boldsymbol{u}-\boldsymbol{u}_g\right)^T R\left(\boldsymbol{u}-\boldsymbol{u}_g\right)$
  - Cost has linear terms
    - Value function takes on the form: quadratic + linear + constant
    - Recursion updates for all terms
    - Controller is affine
- Tracking LQR $(\boldsymbol{x}_t-\widetilde{\boldsymbol{x}}_t)^T Q(\boldsymbol{x}_t-\widetilde{\boldsymbol{x}}_t)+(\boldsymbol{u}_t-\widetilde{\boldsymbol{u}}_t)^T R\,(\boldsymbol{u}_t-\widetilde{\boldsymbol{u}}_t)$
  - Have a desired trajectory to follow
- Iterative LQR
  - Iteratively approximate non-LQR problem as LQR problem and solve several times

Tracking LQR

- Consider desired state and control sequence
  - $(\tilde{\boldsymbol{x}}_0, \tilde{\boldsymbol{x}}_1, \dots, \tilde{\boldsymbol{x}}_{T-1}), (\tilde{\boldsymbol{u}}_0, \tilde{\boldsymbol{u}}_1, \dots, \tilde{\boldsymbol{u}}_{T-1})$
- Consider *error* states, controls, and dynamics
  - $\delta \boldsymbol{x}_t = \boldsymbol{x}_t - \tilde{\boldsymbol{x}}_t$
  - $\delta \boldsymbol{u}_t = \boldsymbol{u}_t - \tilde{\boldsymbol{u}}_t$
  - $\delta \dot{\boldsymbol{x}}_t = \dot{\boldsymbol{x}}_t - \dot{\tilde{\boldsymbol{x}}}_t = A_t \boldsymbol{x}_t + B_t \boldsymbol{u}_t - (A_t \tilde{\boldsymbol{x}}_t + B_t \tilde{\boldsymbol{u}}_t) = A_t \delta \boldsymbol{x}_t + B_{\boldsymbol{t}} \delta \boldsymbol{u}_t$

- Want error → 0
  - Apply standard LQR on *error* states, controls, and dynamics
  - $\delta \boldsymbol{u}_t^* = K_t \delta \boldsymbol{x}_t \Rightarrow \boldsymbol{u}_t^* = K_t \delta \boldsymbol{x}_t + \tilde{\boldsymbol{u}}_t$